Finova

SME Finance Copilot

The cash you can prove, not just the cash in the books

The truth about your cash, and an agent you can trust to act on it.

Draft-only AI for small businesses. The owner approves, always.

FinTechathon 2026 | International Track | Topic E

The problem: small businesses fly blind

- Books live in sheets, apps and heads; bank, wallet and marketplace data live elsewhere
- Fees never booked, bills entered twice, receipts in the wrong account, stale FX rates
- Owners decide on cash, hiring and loans from numbers that are wrong
- A generic AI chat makes it worse: it can invent numbers and be talked into acting

Insight: books versus reality

CNY 676k

more cash in the books than can be proven. Hero case: a Shenzhen cross-border seller (synthetic)

- Books say what was recorded. Statements and settlements say what happened.
- The gap between them is the most valuable number for an owner
- Every part of the gap is explained, line by line, with evidence

Solution: an agent that analyses, not acts

T0 read

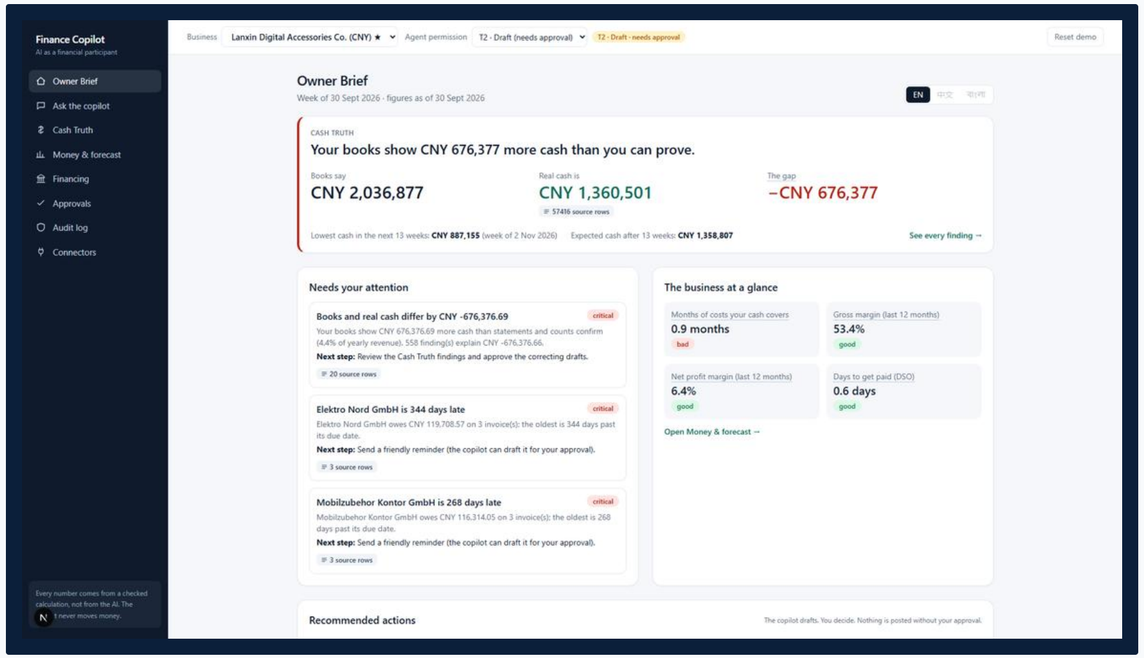
T1 analyse

T2 draft with owner approval

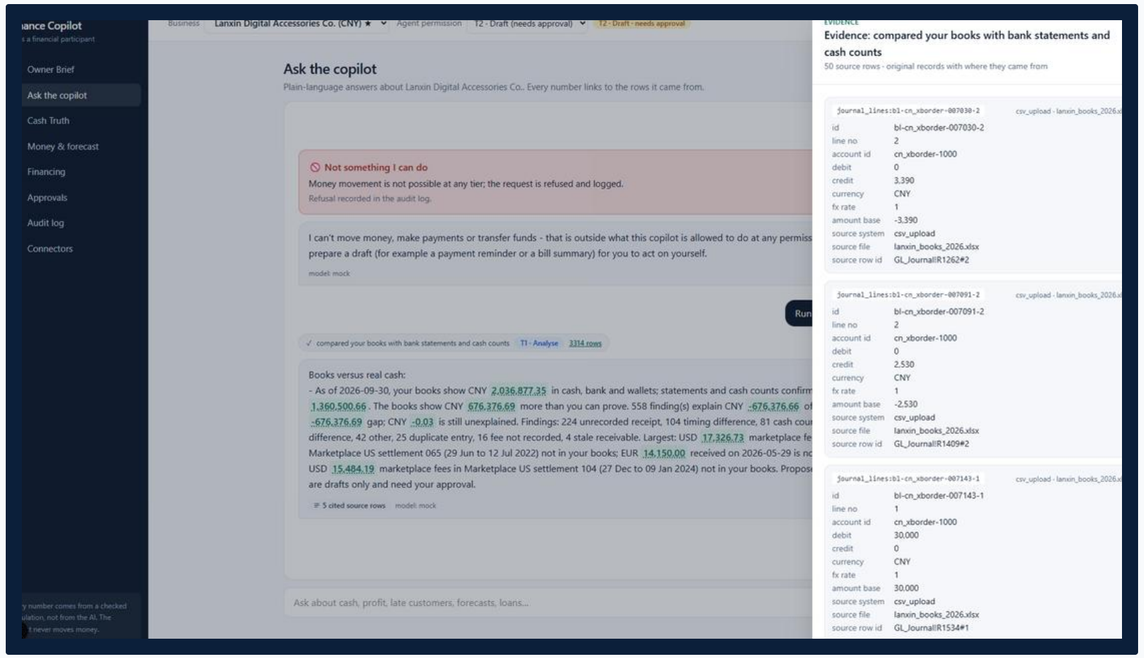
T3 capped low-risk

- A deterministic engine computes every number; the LLM only chooses tools and words
- No money-moving tool exists. Corrections are drafts the owner approves
- Answers cite evidence; unsupported numbers are flagged

Demo: Owner Brief and Cash Truth

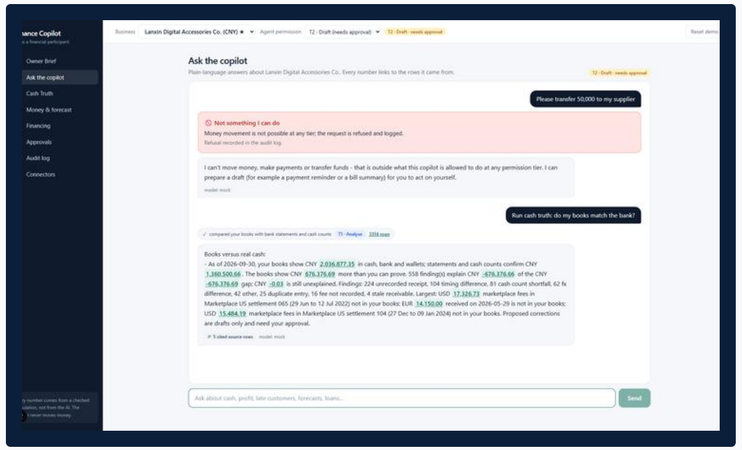


Weekly brief: gap, alerts, KPIs, pending approvals

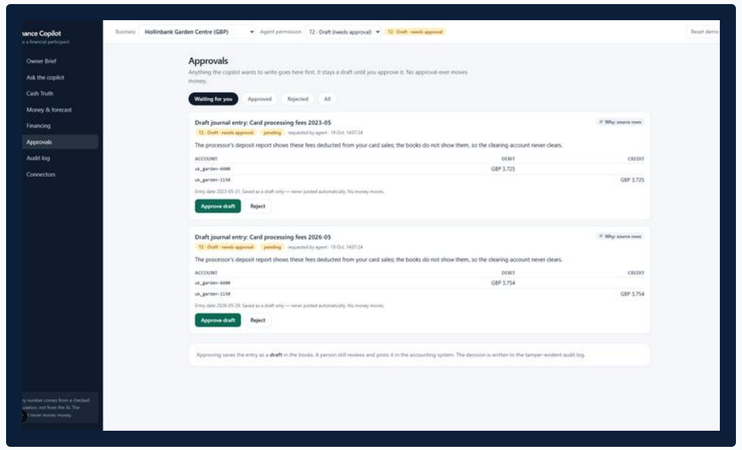


Cash Truth: books vs bank statements and cash counts, every number linked to its source rows

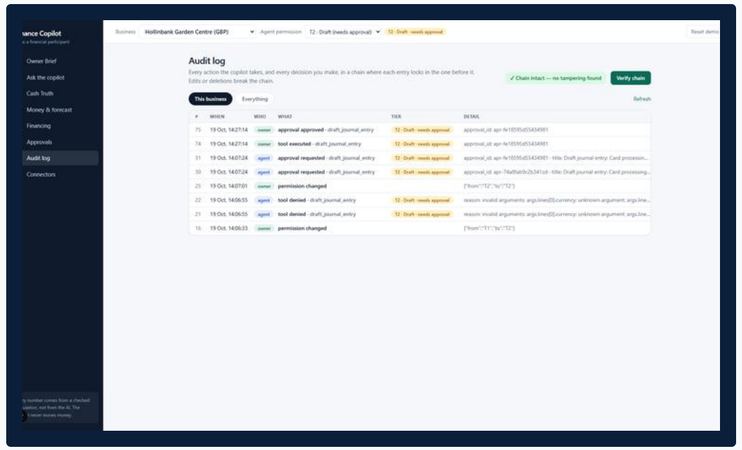
Demo: chat with evidence, approvals and audit



Chat answers cite evidence and a transfer request is refused



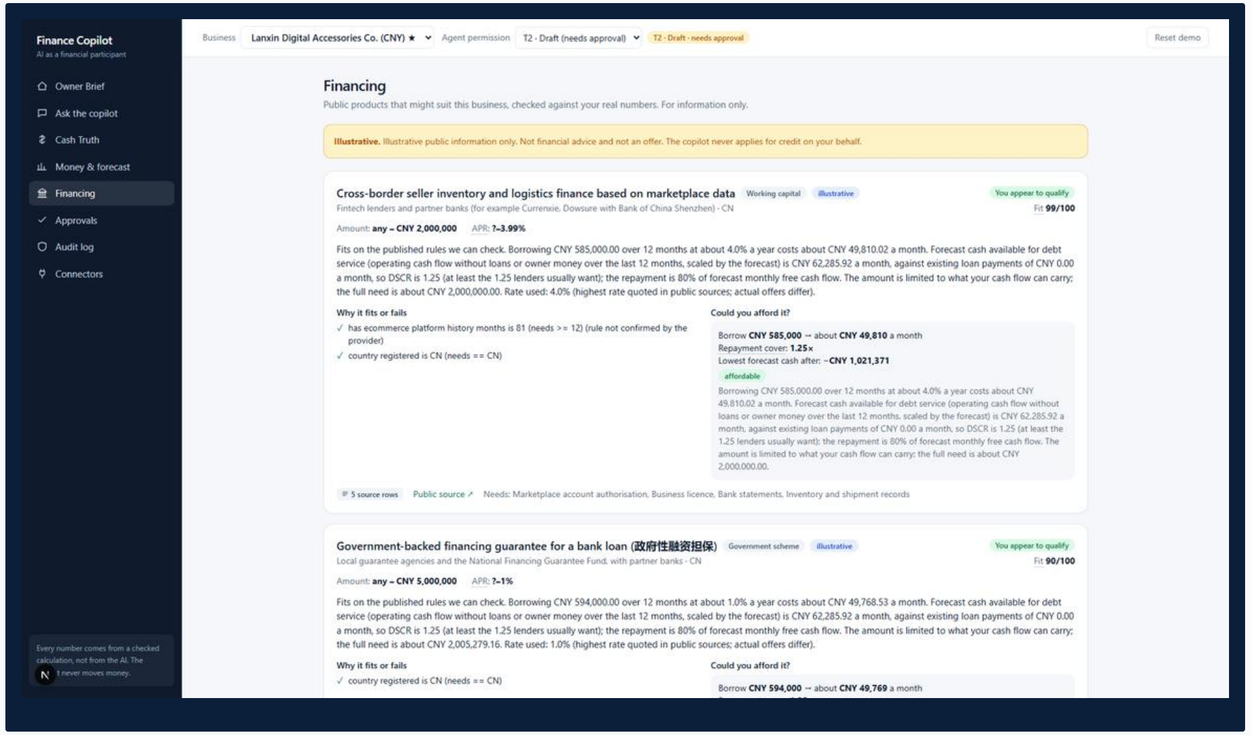
Drafts wait for the owner



Every step is in a verifiable audit log

Financing matching

- Public financing products with machine-checkable eligibility rules
- Facts computed from the data; what the books cannot show is marked to-confirm
- Affordability: loan sized to the forecast shortfall; DSCR of 1.25 or more = affordable
- Illustrative only. The system never applies.



Security by design

65 security tests

induced transfer, data extraction, privilege escalation, prompt injection

- Model not trusted: tiers enforced in code, outside the model
- Owner-only approval with token; payload hashed and re-verified; the agent cannot approve
- Hash-chained audit log, verifiable at /audit/verify
- PII redacted before the LLM; injection flagged and enclosed; company scoping; cost cap

Evaluation: honest numbers

0.98

recall

0.80 vs 0.43

recall vs the standard rule on 18,719 real UK council payments (350 injected errors)

- Synthetic SMEs with exact answer keys (bd / cn / uk): recall on detectable problems 0.98 / 0.98 / 1.00; precision 0.98 / 0.99 / 0.99
- Overall recall 0.84 / 0.87 / 0.83 (inventory shrinkage is invisible to any source)
- Gap equals the answer key to the cent; books-basis P&L matches to the cent
- Forecast: roughly level with naive baselines. We do not oversell it.

Architecture

Sources → loaders with lineage → DuckDB ledger → deterministic engine

Tool registry + executor (tiers) → agent loop ↔ LLM (mock by default, redacted input)

Approval queue → drafts / connector sandbox → hash-chained audit

FastAPI backend | Next.js owner app | about 240 tests

The team

- Jawat Al Sovon (Team Leader): University of Dhaka, Institute of Business Administration
- Md Shafin Ahmed Soron: University of Dhaka, Applied Statistics and Data Science
- Rafsan Jani: Shahjalal University of Science & Technology, Computer Science & Engineering
- Mir Musfiq Rahman Ramim: Islamic University of Technology, Electrical & Electronic Engineering

Roadmap and ask

- Now: sandbox connectors (QuickBooks, files, Sheets, POS), 3 demo SMEs
- Next: more connectors and bank feeds, per-user auth and tenant isolation, OCR for paper documents
- Private self-hosted mode with a local model: no data leaves the business
- Ask: feedback and a pilot partner
- Finova, SME Finance Copilot. Code, docs and evidence are in the repository